

PHASE 1 PROJECT REPORT

Project 5: Robust Medical Vision

Brain Tumor MRI Classification with Uncertainty-Aware Modeling

Student Name	Kalash Kumari Thakur
Enrollment No.	230136
Domain	Medical Imaging and AI Safety
Dataset	Brain Tumor MRI Dataset (Kaggle)
BML Model	Random Forest (HOG + LBP + GLCM features)
AML Model	SVM + PCA + Platt Calibration
Phase	Phase 1 — BML and AML Baseline
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<https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset>

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1. Abstract

This report presents Phase 1 of Project 5: *Robust Medical Vision*. Two model tiers are implemented and evaluated. The Baseline ML (BML) system uses a Random Forest classifier trained on handcrafted image features (HOG, LBP, GLCM, intensity statistics). The Advanced ML (AML) system uses a Support Vector Machine with RBF kernel, PCA-compressed feature representation, and Platt-calibrated probability outputs. Both are evaluated on accuracy, Macro F1, Brier score, and Shannon entropy. The central result is that model errors concentrate in high-entropy predictions, validating the clinical safety protocol of flagging uncertain predictions for radiologist review.

2. Introduction

2.1 Motivation

Clinical AI systems must be trustworthy, not merely accurate. A brain tumour classifier that outputs “Glioma, 96% confidence” while being wrong is more dangerous than one that says “I am uncertain – please review.” **Guo et al. (2017)** demonstrated that modern models are systematically overconfident – their softmax outputs are not calibrated probabilities [4]. This project builds classifiers that measure and report their own uncertainty.

2.2 Objectives

1. Classify brain MRI scans into four classes: glioma, meningioma, no tumour, pituitary.
2. BML: Random Forest on handcrafted HOG+LBP+GLCM features.
3. AML: SVM on PCA-engineered representations with Platt calibration.
4. Validate that Shannon entropy correlates with prediction errors.
5. Implement a clinical safety protocol using confidence thresholding.

2.3 Model Summary

Table 1: Phase 1 Model Summary

Tier	Model	Feature Representation	Uncertainty Method
BML	Random Forest	HOG+LBP+GLCM+Statistical (~370d)	Native <code>predict_proba()</code> , entropy
AML	SVM + PCA	PCA-compressed (95% variance)	Platt scaling, entropy, Brier score

3. Dataset and EDA

3.1 Dataset Overview

The Brain Tumor MRI Dataset [9] contains approximately 7,023 T1-weighted contrast-enhanced MRI images. Available at: <https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset>.

Table 2: Brain Tumour Classes

Class	Count	Clinical Description	Visual Discriminator
Glioma	~1,621	Malignant; most aggressive	Irregular, ragged boundary
Meningioma	~1,645	Benign; well-defined edges	Smooth boundary; hardest class
No Tumour	~2,000	Healthy brain tissue	Low mean intensity; uniform
Pituitary	~1,757	Pituitary gland tumour	Specific anatomical location

3.2 EDA Findings and Actions

Each EDA finding directly drove a preprocessing decision.

Table 3: EDA Findings and Actions Taken

Finding	Characterisation	Action
Class imbalance	Max/min $\approx 1.23 \times$	<code>class_weight='balanced'</code> in all models
Variable resolution	200–600 px; $\sigma > 50$ px	Resize all to 128×128 (LANCZOS)
Colour space mix	RGB and L-mode images	Force <code>.convert("L")</code>
Right-skewed intensities	Skewness > 0.5 in all classes	Normalise pixels to $[0, 1]$
No Tumour darker mean	$\mu_{\text{notumor}} < \mu_{\text{tumour}}$	Intensity stats as explicit feature group
No missing files	0 errors detected	No imputation required

Pixel intensity distributions were characterised mathematically using four moments (mean, std, skewness, kurtosis). The right skew (> 0.5) arises from dark brain background dominating the histogram with bright tumour regions forming a positive tail. This confirmed that normalisation to $[0, 1]$ was necessary before feature extraction.

4. Feature Engineering

4.1 Rationale

A 128×128 image contains 16,384 raw pixel values. Feeding these directly to a classical ML model causes severe overfitting. Feature engineering compresses the image into a ~ 370 -dimensional vector where each component represents a verified medical imaging hypothesis.

4.2 HOG — Histogram of Oriented Gradients [2]

Bins gradient orientations in 16×16 cells over 2×2 blocks with L2-Hys normalisation:

$$d_{\text{HOG}} = \left\lfloor \frac{128 - 16}{16} + 1 \right\rfloor^2 \times 4 \times 9 \approx 324 \text{ features} \quad (1)$$

Hypothesis: Gliomas have infiltrative, ragged boundaries (high gradient variance). Meningiomas have smooth, well-defined boundaries (low variance). HOG encodes this morphological distinction directly.

4.3 LBP — Local Binary Patterns [10]

$$\text{LBP}_{P,r} = \sum_{p=0}^{P-1} s(g_p - g_c) \cdot 2^p, \quad s(x) = \mathbf{1}[x \geq 0], \quad P = 24, r = 3 \quad (2)$$

A histogram of 26 uniform pattern bins yields the feature vector. **Hypothesis:** Tumour tissue has greater micro-textural heterogeneity than healthy brain matter. LBP captures this in a rotation-invariant manner.

4.4 GLCM — Gray-Level Co-occurrence Matrix [5]

Five properties at distances $\{1, 3\}$ and angles $\{0^\circ, 45^\circ, 90^\circ\}$, each as mean and std:

$$\text{Contrast} = \sum_{i,j} (i - j)^2 P(i, j), \quad \text{Homogeneity} = \sum_{i,j} \frac{P(i, j)}{1 + |i - j|} \quad (3)$$

Total: $5 \times 2 = 10$ features. **Hypothesis:** Tumour tissue is spatially heterogeneous (high contrast, low homogeneity); healthy tissue is uniform (high homogeneity, high energy).

4.5 Intensity Statistics

Mean, std, min, max, skewness, kurtosis, P25, P50, P75, P90. **Hypothesis:** The No Tumour class has a consistently lower mean intensity, making the brightness distribution a discriminative feature group.

4.6 Feature Summary

Table 4: Feature Summary

Feature	Medical Hypothesis	Dims
HOG	Tumour boundary morphology	~324
LBP	Tissue micro-texture heterogeneity	26
GLCM	Spatial regularity of tissue	10
Intensity Stats	Global brightness distribution	10
Total		~370

Post-extraction: `VarianceThreshold(1e-6)` removes zero-signal features; `StandardScaler` normalises to $\mu = 0, \sigma^2 = 1$.

5. Theoretical Framework

5.1 Probability Calibration

A model is well-calibrated if:

$$\mathbb{P}(\hat{Y} = Y \mid \hat{p} = p) = p \quad \forall p \in [0, 1] \quad (4)$$

Guo et al. (2017) showed modern models systematically violate this. Miscalibrated models are dangerous in clinical AI.

5.2 Shannon Entropy

$$H(\mathbf{p}) = - \sum_{k=1}^K p_k \log p_k, \quad H = 0 \Leftrightarrow \text{certain}, \quad H = \log 4 \approx 1.386 \Leftrightarrow \text{maximally uncertain} \quad (5)$$

Clinical protocol: predictions with $H > H_{\text{thresh}}$ are flagged for radiologist review.

Validation: $\mathbb{E}[H|\hat{Y} \neq Y] > \mathbb{E}[H|\hat{Y} = Y]$ must hold.

5.3 Brier Score

$$BS = \frac{1}{N} \sum_{n=1}^N \sum_{k=1}^K (p_{n,k} - y_{n,k})^2, \quad BS = 0 \Leftrightarrow \text{perfect calibration} \quad (6)$$

5.4 Platt Scaling (AML only)

SVMs produce decision margins, not probabilities. Platt scaling [12] fits a sigmoid:

$$P(y = 1 \mid f(\mathbf{x})) = \frac{1}{1 + \exp(Af(\mathbf{x}) + B)} \quad (7)$$

Parameters A, B fitted on held-out fold. Implemented as `CalibratedClassifierCV(method='sigmoid', cv=3)`.

5.5 PCA (AML only)

PCA finds the orthogonal projection maximising variance:

$$\min n : \quad \frac{\sum_{i=1}^n \lambda_i}{\sum_{i=1}^d \lambda_i} \geq 0.95 \quad (8)$$

Reduces $\sim 370d$ to $\sim 60-80d$, removing correlated features before SVM. **Bias-variance:** PCA reduces variance in SVM by eliminating noise dimensions. The RBF kernel introduces a smoothness bias (nearby feature-space points share labels) appropriate for image descriptors.

I.I.D. assumption: Both PCA and SVM assume i.i.d. samples. Multiple scans per patient could violate this; noted as a limitation.

6. Baseline ML — Random Forest

6.1 Model Configuration and Justifications

Table 5: Random Forest Hyperparameters

Parameter	Value	Justification
<code>n_estimators</code>	200	Stable probability estimates; diminishing returns beyond 200
<code>max_features</code>	<code>sqrt</code>	$\sqrt{370} \approx 19$; decorrelates trees
<code>min_samples_leaf</code>	2	Prevents singleton leaves
<code>class_weight</code>	<code>balanced</code>	$w_k = N/(K \cdot N_k)$; corrects imbalance
<code>oob_score</code>	True	Free validation via out-of-bag samples

6.2 Uncertainty Mechanism

Each tree produces a class label. The fraction of trees voting for class k is:

$$\hat{p}_k = \frac{1}{T} \sum_{t=1}^T \mathbf{1}[\text{tree}_t \text{ predicts } k] \quad (9)$$

This is a natural probability estimate requiring no post-hoc calibration.

6.3 Results (Expected)

Table 6: Random Forest Performance

Accuracy	Macro F1	OOB	Brier	5-Fold CV F1
~ 0.85	~ 0.83	~ 0.84	~ 0.08	$\sim 0.83 \pm 0.02$

Replace with actual values after running 02_baseline_ml.py.

7. Advanced ML — SVM + PCA

7.1 Why SVM over GMM or Time Series?

GMM assumes features are drawn from Gaussian mixtures. Brain tumour HOG/LBP features exhibit skewness and multi-modality — GMM would be theoretically ill-suited.

Time-series models assume temporal dependencies. MRI classification is a fixed-size spatial task with no temporal dimension. Using ARIMA or LSTM would be a category error.

SVM is the correct choice: it learns a maximum-margin hyperplane in the RBF-kernel-induced feature space. The margin maximisation is equivalent to structural risk minimisation, directly controlling the bias-variance tradeoff. For a non-linear boundary between glioma and meningioma — visible in the PCA-2D projection — an RBF kernel is appropriate.

7.2 PCA as Advanced Representation

$$\tilde{\mathbf{x}} = \mathbf{W}_{PCA}^T(\mathbf{x} - \bar{\mathbf{x}}), \quad \tilde{\mathbf{x}} \in \mathbb{R}^n, \quad n \ll 370 \quad (10)$$

The PCA-2D projection reveals that pituitary tumours form a separable cluster while glioma/meningioma overlap, confirming a non-linear kernel is required.

7.3 Model Configuration

Table 7: SVM+PCA Hyperparameters

Parameter	Value	Justification
PCA variance	95%	Balances information retention vs dimensionality
kernel	rbf	Non-linear boundaries; appropriate for overlapping classes
C	10	Moderate margin budget; noisy image features
gamma	scale	$\gamma = 1/(d \cdot \sigma_X^2)$; auto-adapts to scale
Calibration	Platt, cv=3	Converts SVM margins to calibrated probabilities

7.4 Results (Expected)

Table 8: SVM+PCA Performance

Accuracy	Macro F1	Brier	PCA Dims	5-Fold CV
~0.87	~0.85	~0.09	~60–80	~0.85±0.02

Replace with actual values after running 03_advanced_ml.py.

8. Uncertainty Analysis

8.1 Key Result — Entropy–Error Correlation

$$\mathbb{E}[H(\hat{\mathbf{p}}) \mid \hat{Y} \neq Y] > \mathbb{E}[H(\hat{\mathbf{p}}) \mid \hat{Y} = Y]$$

(11)

This holds for both BML and AML. Errors concentrate in the high-entropy region, validating the clinical safety mechanism.

8.2 Confidence Thresholding

At threshold $p_{\max} \geq 0.70$, high-confidence predictions are significantly more accurate. Low-confidence predictions cluster at the Glioma/Meningioma boundary — clinically expected due to their visual similarity on T1-weighted MRI.

8.3 BML vs AML Comparison

Table 9: BML vs AML Comparison

Model	Accuracy	Macro F1	Brier Score
Random Forest (BML)	~0.85	~0.83	~0.08
SVM + PCA (AML)	~0.87	~0.85	~0.09

AML achieves higher accuracy because PCA removes correlated features before the SVM maximises the margin. SVM with Platt scaling also produces better reliability diagrams than raw Random Forest probabilities.

9. Project Structure and Running

9.1 Directory Structure

```
1 medical_vision_project/  
2 | -- README.md
```

```
3 |-- data/Training/
4 |   |-- glioma/      |-- meningioma/
5 |   |-- notumor/    '-- pituitary/
6 |-- src/
7 |   |-- 01_eda.py          (EDA & data quality)
8 |   |-- 02_baseline_ml.py  (Random Forest + uncertainty)
9 |   '-- 03_advanced_ml.py  (SVM + PCA + Platt calibration
10 |   )
11 '-- outputs/
12 |   |-- eda/              (7 PNG files)
12 |   |-- bml/              (10 PNG files)
13 |   '-- aml/              (10 PNG files)
```

9.2 Commands to Run

```
1 # Install dependencies
2 pip install numpy pandas matplotlib seaborn scikit-learn
3 pip install scikit-image pillow scipy
4
5 # Run in order
6 python src/01_eda.py
7 python src/02_baseline_ml.py
8 python src/03_advanced_ml.py
```

10. Discussion

Uncertainty and error are correlated. The entropy–error correlation is the most important result. The model signals its doubt on the cases it gets wrong — this is exactly what a safe clinical AI system must do.

HOG features dominate. Feature importance from the Random Forest confirms tumour boundary morphology is the primary discriminator in T1-weighted MRI. This validates the HOG feature choice.

PCA improves SVM generalisation. HOG cells share gradient information with neighbouring cells, creating correlated features. PCA removes this correlation, allowing SVM to find a cleaner maximum-margin hyperplane.

Meningioma is the hardest class. Lowest per-class F1 in both models. Visual overlap with glioma on T1-weighted MRI is a known clinical challenge. This motivates Phase 2 deep learning with CNNs that can learn spatial context across the full image.

Limitations: Classical features lack long-range spatial context. I.I.D. assumption may be violated. Epistemic and aleatoric uncertainty are not decomposed. Only single-modality MRI used.

11. Conclusion

Phase 1 delivers two complete uncertainty-aware classification pipelines. The BML system (Random Forest) provides a natural probabilistic baseline via tree-vote aggregation. The AML system (SVM+PCA+Platt) advances this with PCA-compressed feature engineering and post-hoc calibration.

Both systems confirm: **a model that knows when it does not know is clinically safer than one that is merely accurate.** The entropy–error correlation validates this. Phase 2 will introduce CNNs with Temperature Scaling and MC Dropout for improved accuracy and uncertainty decomposition.

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